



Breast Cancer Subtyping by Surface-Enhanced Raman Scattering (SERS) Signature of Plasma-Derived Extracellular Vesicles

Fatemeh Pashizeh¹ · Anastasiia Tukova¹ · Simon Chang-Hao Tsao^{1,2} · Victoria Ie Ching Tan¹ · Dhanusha Sabanathan³ · Su Su Thae Hnit¹ · Yuling Wang¹

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Abstract

Breast cancer subtype classification has critical implications for patients' treatment selection and disease prognosis. Small extracellular vesicles (sEVs) have emerged as promising biomarkers for liquid biopsy due to their ability to reflect the molecular characteristics of their parent cells and are found in high concentration in biological fluids, such as plasma and urine. However, accurately distinguishing breast cancer subtypes using sEVs remains challenging because of the subtle biochemical differences among heterogeneous sEV populations. In this study, label-free surface-enhanced Raman scattering (SERS) was employed to analyze plasma-derived sEVs as a comprehensive molecular profiling approach, capable of capturing biochemical information without prior target selection or labelling. To extract discriminative features from complex spectral datasets among sEV populations, label-free SERS was combined with a support vector machine (SVM) classifier, establishing a precise, data-driven framework for molecular subtyping. The integrated SERS-SVM platform successfully differentiated human epidermal growth factor receptor 2 negative (HER-2^{neg}) breast cancer, HER-2 positive (HER-2^{pos}) breast cancer, and healthy controls, highlighting its potential as a rapid, non-invasive, and high-throughput liquid biopsy approach for breast cancer diagnosis and molecular subtyping. In our study, the SVM model achieved an overall accuracy of 95% in internal train–test validation and 85% in an independent validation using a separate set of clinical samples. Importantly, the independent test set showed 100% sensitivity for both HER-2^{neg} and HER-2^{pos} breast cancer groups, indicating that the model correctly identified all breast cancer subtype samples in the validation cohort.

Keywords Breast cancer · Extracellular vesicles (EVs) · Human epidermal growth factor receptor 2 (HER-2) · Label-free surface-enhanced Raman scattering · Machine learning · Support vector machine (SVM)

1 Introduction

Breast cancer is the most frequently diagnosed malignancy among women globally and continues to represent a significant public health issue. Breast cancer represents nearly one-third of all cancer diagnoses in women, and its global burden is expected to rise by 40% by the year 2040 [1]. Breast cancer can be classified into different subgroups based on molecular features of cancer tissue. This classification is essential for therapy selection because each subtype is associated with specific therapeutic strategies that influence treatment response [2]. One of the most important molecules in breast cancer classification is the human epidermal growth factor receptor 2 (HER-2), which is a member of the tyrosine kinase family. It is encoded by the *erb-b2* receptor tyrosine kinase 2 (*ERBB2*) proto-oncogene located on chromosome 17q12 [3]. HER-2 activates downstream

Fatemeh Pashizeh and Anastasiia Tukova have contributed equally to this work.

✉ Su Su Thae Hnit
susuthae.hnit@mq.edu.au

✉ Yuling Wang
yuling.wang@mq.edu.au

¹ School of Natural Sciences, Faculty of Science and Engineering, Macquarie University, Sydney 2109, Australia

² Department of Surgery, Austin Health, Melbourne 3084, Australia

³ Macquarie Medical School, Faculty of Medicine, Health and Human Sciences, Macquarie University, Sydney 2109, Australia

signalling pathways involved in cell proliferation, invasion, angiogenesis, and resistance to apoptosis [3]. HER-2 overexpression is observed in approximately 15%–20% of breast cancers [3], and its expression is evaluated by both protein expression and gene amplification [2]. According to the Royal College of Pathologists of Australasia (RCPA) breast cancer molecular subtyping guidelines, clinical pathology assessment, including immunohistochemistry (IHC) and *in situ* hybridization (ISH) for confirmation of HER-2 gene amplification cases, is classified as HER-2 negative (HER-2^{neg}; IHC 0, IHC 1⁺, and IHC 2⁺ and ISH-non-amplified) and HER-2 positive (HER-2^{pos}; IHC 2⁺ and IHC 3⁺, ISH amplified) [3, 4]. This classification of HER-2^{neg} and HER-2^{pos} guides therapy selection and influences clinical outcomes, as tumours expressing HER-2 are sensitive to HER-2-targeted therapies, including monoclonal antibodies such as Trastuzumab, as well as small-molecule inhibitors such as Lapatinib [5].

Current molecular subtyping methods rely on tumour tissue biopsy or surgical resection, which are invasive procedures and unable to comprehensively capture the dynamic molecular heterogeneity of tumours, restricting their application in dynamic monitoring of disease [6]. Therefore, there is an urgent need for effective, low-risk, and non-invasive approaches.

Liquid biopsy offers a minimally invasive approach to identify tumour-associated biomarkers in body fluids such as blood, facilitating dynamic monitoring of patient progression and therapeutic response. Extracellular vesicles (EVs) are nanosized lipid bilayer particles released by all cell types. They carry various biomolecular cargo, including proteins, nucleic acids, and lipids, present on their surface or loaded within them [7]. In cancer, EVs mediate intercellular communication by transferring oncogenic factors, thereby promoting tumour progression, metastasis, immune evasion, and drug resistance [8, 9]. EVs are abundantly present in biological samples and contain molecular signatures that reflect those of their parent cells. EVs are categorized into small EVs (sEVs; < 200 nm) and medium/large EVs (> 200 nm) [10]. sEVs originate from the endosomal pathway, enabling the selective packaging of a wide variety of cargoes [11, 12]. For example, they carry transmembrane proteins such as CD9, CD63, and CD81 as well as tumour-specific markers such as epithelial cell adhesion molecule (EpCAM) and immune checkpoints such as programmed death-1 (PD-1) [8].

Surface-enhanced Raman scattering (SERS) is a spectroscopic analytical technique for the ultrasensitive detection of trace amounts of analytes and has shown great potential for biomarker detection [13]. Under optimized conditions, SERS can achieve up to 10⁶ signal amplification, generating a unique molecular vibrational spectrum [14]. Two

measurement strategies are commonly used in SERS: label-based SERS and label-free SERS. Similar to conventional techniques, label-based SERS requires the identification of specific biomarkers for targeted detection. In contrast, label-free SERS can capture molecular vibrational signals from all molecules present in sEVs, including both known and unknown biomarkers, providing valuable biochemical information [15]. However, due to the complex biomolecular composition of the sEV population, which contains diverse proteins, lipids, and cancer-associated antigens, label-free SERS generates spectra that are convoluted and difficult to analyse and interpret without sophisticated mathematical processing [16, 17]. To overcome this limitation, machine learning algorithms and multivariate statistical approaches were integrated with SERS to identify and enhance subtle compositional differences in sEVs' molecular profiles [17–19]. This combined strategy offers a data-driven framework for identifying spectroscopic variations that are difficult to detect using traditional spectral analysis, thereby offering a reliable, non-invasive, and high-throughput platform for cancer subtyping and classification [15]. For example, recent studies on breast cancer [19] and lung cancer [20] demonstrated the potential application of label-free SERS for cancer subtyping and diagnosis [20].

In this study, we aimed to evaluate the ability of label-free SERS to distinguish sEVs isolated from the plasma of healthy donors and patients with breast cancer, as well as to differentiate between HER-2^{neg} and HER-2^{pos} breast cancer subtypes. It was not intended to distinguish breast cancer from other cancer types. Rather, the aim was to classify breast cancer according to its HER-2 molecular subtype rather than disease stage, thereby providing a potential tool to support HER-2-guided therapeutic decision-making.

To preserve the biological integrity of sEVs and minimise sample damage, this study introduces a liquid-based, label-free SERS approach for breast cancer HER2 subtyping. We employed a capillary-based liquid SERS platform instead of the conventional drop-cast approach, as drying-induced effects can alter the structural conformation of biomolecules and influence spectral features. For data analysis, principal component analysis (PCA) was used to reduce the dimensionality of the SERS dataset and assess spectral variation, while a support vector machine (SVM) algorithm was applied for machine learning-based classification and prediction. The results demonstrate that this label-free SERS approach enables accurate identification of plasma-derived sEVs from healthy and breast cancer samples and provides the capability to discriminate between HER-2^{neg} and HER-2^{pos} breast cancer.

2 Experimental

2.1 Materials

The MCF-7, SKBR-3, and MCF-10A cell lines were purchased from American Type Culture Collection (ATCC). Roswell Park Memorial Institute 1640 (RPMI 1640; Cat No. R8758), McCoy's 5A (modified) medium (Cat No. 16600-082), Dulbecco's modified Eagle medium/nutrient mixture F-12 (DMEM/F-12; Cat No. 10565018), human epidermal growth factor (EGF) recombinant protein (Cat No. PHG0311L), chlorotoxin (CAS No. 9012-63-9), insulin (Cat No. I0516), hydrocortisone (CAS No. 50-23-7), horse serum (HS; Cat No. 16050122), fetal bovine serum (FBS; Cat No. 10099141), TrypLE Express enzyme (Cat No. 12604021), penicillin–streptomycin (P/S; Cat No. P4333), Dulbecco's phosphate buffered saline (DPBS; Cat No. D8537), anti-CD9 Alexa Fluor® 647 (Clone MEM-61, Abcam, Cat No. ab187776), Di-8-ANEPPS (CAS No. 157134-53-7), qEV original columns (Izon, Cat No. ICO-35), chloroauric acid (HAuCl₄; CAS No. 16961-25-4), trisodium citrate tribasic dihydrate (C₆H₅Na₃O₇·2H₂O; CAS No. 61332-01-3), potassium iodide (KI; CAS No. 7681-11-0), and ultra-pure water (Millipore Milli-Q grade) with a resistivity of 18.2 MΩ·cm were used in all the experiments.

2.2 Cell Culture

The human breast cancer cell lines MCF-7 and SKBR-3, and the non-tumorigenic breast epithelial cell line MCF-10A, were cultured under standard conditions at 37 °C in a humidified atmosphere containing 5% CO₂. MCF-7 cells were maintained in RPMI 1640 medium supplemented with 10% FBS and 1% P/S. SKBR-3 cells were cultured in McCoy's 5A medium supplemented with 10% FBS and 1% P/S. MCF-10A cells were maintained in DMEM/F12 medium supplemented with 5% HS, 10 µg/mL insulin, 0.5 µg/mL hydrocortisone, 20 ng/mL EGF, 100 ng/mL cholera toxin, and 1% P/S.

When cell confluence reached approximately 80%, cells were washed twice with 10 mL of DPBS to remove residual FBS and then incubated in FBS-free culture medium for 48 h. After the incubation period, the conditioned medium containing EV was collected for sEV isolation and downstream experiments.

2.3 Isolation of Cell Line-Derived sEVs

sEVs were isolated from FBS-free conditioned medium using a combination of differential centrifugation, ultracentrifugation (UC), and size-exclusion chromatography (SEC). A total of 400 mL of conditioned medium was

collected from MCF-10A cells, and 200 mL from MCF-7 and SKBR-3 cells. Briefly, the collected conditioned medium was sequentially centrifuged at 500 g and 2000 g for 10 min to remove detached cells and cell debris, followed by filtration through a 0.2 µm filter to eliminate large particles. Then, the supernatant was ultracentrifuged at 120,000 g for 75 min at 4 °C to concentrate the conditioned medium. The final pellet was resuspended in 500 µL DPBS and subjected to SEC for sEV isolation, and five fractions (0.5 mL each) were collected using the SEC isolation method. Then, the first three fractions were concentrated by UC at 120,000 g for 75 min at 4 °C. Finally, the resulting pellet was resuspended in 50 µL of DPBS and stored at −30 °C until further analysis.

2.4 Clinical Samples and Ethical Statement

Plasma samples were collected from 14 healthy donors, 14 patients with HER-2^{neg} breast cancer, and 14 patients with HER-2^{pos} breast cancer. Clinical samples were obtained from different centers, including blood from healthy donors collected by the Australian Red Cross Lifeblood, breast cancer patient samples obtained through the Macquarie University Cancer Biobank, Sydney, Australia, and the St Vincent's Hospital Biobank, Melbourne, Australia. All procedures were conducted in accordance with the guidelines of the Human Research Ethics Committee and the Australian National Statement on Ethical Conduct in Human Research. The study protocol was approved by the Macquarie University Human Research Ethics Committee (Reference No. 52026923066675). Written informed consent was obtained from all participants, including healthy donors and breast cancer patients, prior to the collection and use of their biological samples for research purposes.

2.5 Isolation of Plasma-Derived sEVs

Plasma samples obtained from healthy individuals and patients with HER-2^{neg} or HER-2^{pos} breast cancer were used for sEV isolation using three different methods: UC, SEC–UC, and a combined UC–SEC–UC method. Briefly, 100 µL of plasma was centrifuged at 10,000 g for 30 min at 4 °C to remove debris and larger particles. The resulting supernatant was subsequently subjected to one of the following isolation techniques. For the UC method, the supernatant was ultracentrifuged at 120,000 g for 75 min at 4 °C. The supernatant was discarded, and the pellet was washed by resuspension in 1 mL DPBS, followed by a second UC at 120,000 g for 75 min at 4 °C. The final pellet was then resuspended in 50 µL DPBS. For SEC–UC isolation, after centrifuging to remove large vesicles, the plasma sample was diluted with DPBS to reach 500 µL and then inserted into qEV original columns

(Izon Science, USA). The first three fractions collected from the SEC column were pooled and concentrated by UC at 120,000 *g* for 75 min at 4 °C. The resulting pellet was resuspended in 50 µL of DPBS. For the UC–SEC–UC method, sEVs initially isolated by UC were subsequently purified using SEC with qEV original columns. The first three fractions collected from the SEC column were concentrated by an additional UC step at 120,000 *g* for 75 min at 4 °C. The final pellet was resuspended in DPBS. All isolated sEV samples were stored at –30 °C until further analysis.

2.6 Characterisation of the Isolated sEVs

2.6.1 Nano-Flow Cytometry (NanoFCM)

NanoFCM (NanoAnalyzer U30, NanoFCM, China) was used to determine the size, concentration, and surface marker expression of sEVs. Prior to sample analysis, the instrument was calibrated according to the manufacturer's instructions using 250 nm quality control beads for concentration standardization and monodisperse silica nanoparticles (68, 91, 113, and 155 nm) for size calibration. All procedures were conducted according to the manufacturer's recommendations and established protocols. Data were analyzed using NF Profession software (version 2.3).

For surface marker analysis, 50 µL sEV samples (~ 10⁹ particles/mL) were incubated with fluorophore-conjugated antibodies at 37 °C for 30 min with shaking at 300 r/min using a Thermomixer (Eppendorf, Germany).

2.6.2 Transmission Electron Microscopy (TEM)

For electron microscopy (EM) analysis, sEVs were negatively stained with uranyl acetate. 5 µL of the enriched sEV suspension was applied onto a Formvar/carbon-coated copper grid and incubated for 5 min to allow adsorption. Excess liquid was removed using filter paper, and the loading step was repeated twice to enhance particle deposition. Subsequently, the grid was stained with 10 µL of uranyl acetate replacement stain (ProSciTech) for 10 min. After careful blotting to remove excess stain, the grid was air-dried overnight at room temperature. Imaging was performed using a transmission electron microscope (JEOL, Japan).

2.7 Synthesis of 50 nm Gold Nanoparticles (AuNPs)

AuNPs (~ 50 nm) were synthesized using the citrate reduction based on the Turkevich method [21]. Chloroauric acid (HAuCl₄, 0.5 mL, 1% weight/volume (*w/v*)) was diluted with 49.5 mL of Milli-Q water (18.2 MΩ·cm at 25 °C, Milli-Q Direct water purification system, Merck, Australia) in a 100 mL Erlenmeyer flask, which was covered with

aluminium foil. The mixture was heated on a hotplate at 250 °C and 600 r/min under continuous stirring. Once boiling had been achieved, 0.45 mL of sodium citrate (Na₃C₆H₅O₇, 1% *w/v*), was rapidly added to the solution. The reaction mixture was maintained under heating and stirring for 20 min, followed by the heating being switched off and an additional 10 min of stirring. Subsequently, the solution was allowed to cool to room temperature and stored at 4 °C. Further, KI was used to displace citrate on the AuNP surface to prevent spectral background from the plasmonic substrate. 1 mL of AuNPs was centrifuged at 5500 r/min for 8 min. The supernatant was removed, and the pellet was resuspended in 50 µL of Milli-Q water. After that, we added 10 µL of 1 mM KI solution to the resuspended pellet and incubated it for 15 min before using the nanoparticles.

2.8 Characterisation of AuNPs

2.8.1 TEM of AuNPs

TEM was performed using a field-emission TEM (2100F, JEOL, Japan) operated at an accelerating voltage of 100 kV and 150,000× magnification to evaluate the size and morphology of the nanoparticles. For TEM grid preparation, 10 µL of the nanoparticle suspension was placed onto a carbon-coated copper grid and allowed to stand for 10 min to enable particle adsorption. Excess liquid was carefully removed using filter paper, and the grid was subsequently air-dried overnight before imaging.

2.8.2 Size and Surface Charge Characterization of AuNPs

The hydrodynamic size and surface charge (zeta potential) of the AuNPs were characterized by dynamic light scattering (DLS) using a Zetasizer Nano ZS instrument (Malvern Panalytical, UK) at 25 °C.

2.8.3 UV–Visible (UV–Vis) Spectroscopy

The extinction spectra of AuNPs were measured using a UV–vis spectrophotometer (Jasco V-760, Jasco, Japan) over a wavelength range of 400–900 nm. For analysis, 500 µL of the sample solution was diluted with 500 µL of Milli-Q water and transferred into a disposable 1 mL cuvette (18B/Q/10, Starna, UK).

2.9 SERS Measurement

An upright confocal Raman microscope (alpha300R, WITec GmbH, Germany) with a 10× objective lens was used for SERS measurements. The sample was excited with a 633 nm laser with the power set to 36 mW and an integration

time of 20 s per spectrum. For sample preparation, 10 μL of sEV sample was mixed with 5 μL of AuNPs. For each sample, including both cell line- and plasma-derived sEVs, spectra were collected in 3 biological replicates with 30 measurement points each.

2.10 PCA

SERS spectra pre-processing and PCA were done using OriginLab software (version 2025, OriginLab Corporation, USA). Raw spectra were selected between 400 and 1800 cm^{-1} , smoothed with a Savitzky–Golay filter (8 pt, data step $\sim 1 \text{ cm}^{-1}$), and baselined using the asymmetric least squares (ALS) method. Spectra were then normalised from 0 to 1 and used for PCA analysis.

2.11 Machine Learning

Data processing with SVM was performed in OriginLab as previously described (region-cropped, smoothed, baseline corrected, and normalised), and analysis was conducted with the Quasar package distribution of Orange: Data Mining Toolbox (Bioinformatics Lab, University of Ljubljana) [22]. Processed Raman spectra were classified using a radial basis function support vector machine (RBF-SVM). Each spectrum was treated as a single observation. Raman-shift intensities were used as predictor variables, while cell-line identity was used as the target class. The RBF-SVM model was configured with a cost parameter of $C = 3$ and

$\gamma = 0.00123$, corresponding to the inverse of the number of spectral variables (for plasma samples).

3 Results and Discussion

3.1 Workflow

In this study, plasma samples were collected from healthy donors, HER-2^{neg} breast cancer patients, and HER-2^{pos} breast cancer patients. Plasma-derived sEVs were isolated using an SEC–UC workflow to enrich the vesicle fraction while reducing soluble protein and plasma-derived contaminants. For sample preparation for the SERS measurement, AuNPs were mixed with the isolated plasma-derived sEVs (Fig. 1a). SERS spectra were then acquired directly in liquid using capillary-based measurements. The obtained spectra were analysed using SVM modelling for classification and prediction (Fig. 1b).

3.2 Characterization of AuNPs

For the SERS experiments, we used 50 nm AuNPs as substrates to enhance the Raman signal. AuNPs were synthesized using the Turkevich method [21]. Briefly, citrate was used as a reducing and capping agent to facilitate the formation and stabilization of AuNPs. The colour change of the solution to ruby red indicated the successful reduction of gold ions into AuNPs [23]. Further, before use, nanoparticles

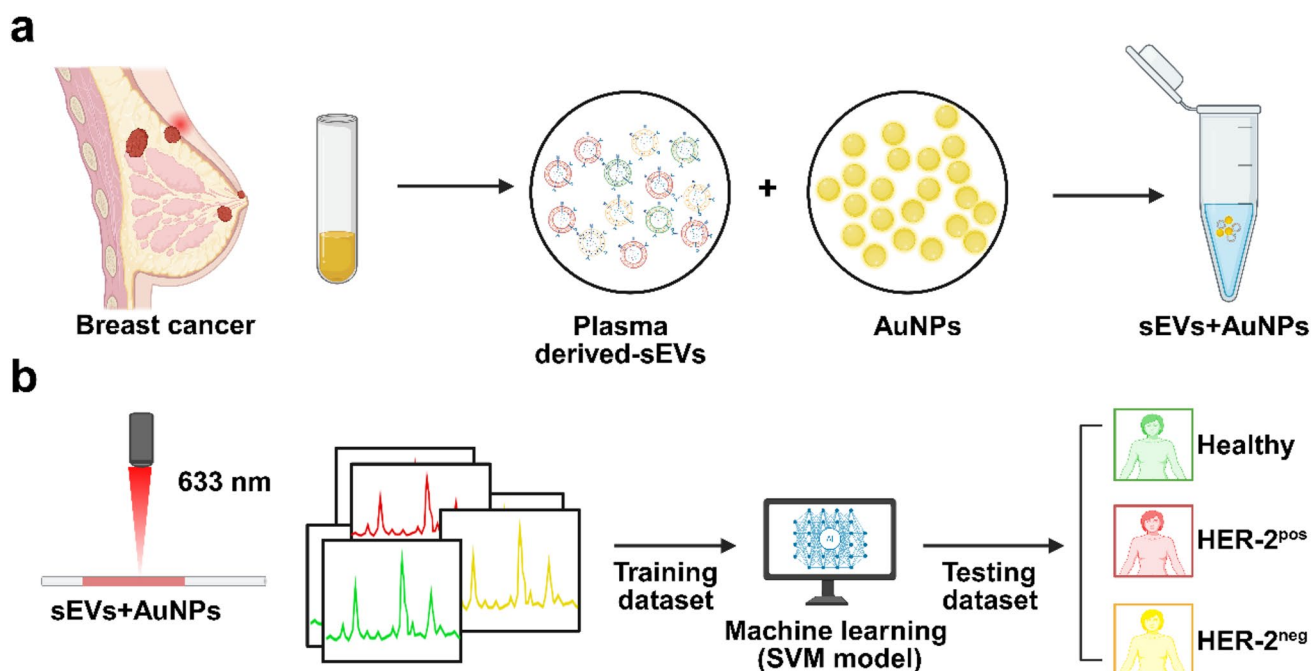


Fig. 1 Schematic workflow of study. **a** Plasma-derived sEVs mixed with AuNPs for SERS measurement. **b** SERS measurement of sEVs using a 633 nm laser in a capillary-based liquid platform, followed by an SVM model for breast cancer diagnosis and molecular subtyping

were incubated with 1 mM KI to replace citrate to prevent its signal from appearing during SERS measurements.

TEM and DLS were used to characterize the size and morphology of the AuNPs. The hydrodynamic diameter of the AuNPs was 48 ± 2 nm. TEM image confirmed their nanospherical morphology and mean size of approximately 50 nm (Fig. S1). UV–vis spectroscopy was used to evaluate the optical characterization of the AuNPs. The AuNP50 exhibited a characteristic surface plasmon resonance (SPR) peak at 538 nm, confirming the successful synthesis and formation of the AuNPs (Fig. S2).

Zeta potential analysis was used to determine the surface charge of the AuNPs, which was found to be -30.8 ± 2.2 mV, indicating that the nanoparticles are well-dispersed and highly stable.

3.3 Isolation Method Selection

The presence of co-isolated contaminants, such as lipoproteins, can confound spectral acquisition and compromise diagnostic accuracy [24, 25]. Previous studies have demonstrated that the purity and heterogeneity of sEVs can vary considerably depending on the isolation method used. As the label-free SERS technique is sensitive to sample purity, we first evaluated the compatibility of isolation methods with SERS measurements [24]. In this study, we employed three different isolation methods, including UC, SEC–UC, and UC–SEC–UC.

To determine the most suitable isolation method for subsequent experiments, SERS spectra were collected from sEVs isolated from a single healthy plasma sample, with three technical replicates. As shown in Fig. 2a, average normalised SERS spectral profiles were observed among sEVs

isolated using the three different methods. UC-isolated samples exhibited higher Raman signal intensities than the SEC–UC and UC–SEC–UC samples. However, PCA further demonstrated that spectra from UC-isolated samples exhibited a broader distribution and reduced overlap between technical replicates, indicating greater heterogeneity within the samples (Fig. 2b). This variability may be attributed to the presence of co-isolated proteins and aggregated vesicles, which can affect spectral consistency [26]. In contrast, more clustered and overlapping spectra indicate improved homogeneity in SEC–UC isolations [27]. Although the SEC–UC method produced an intermediate SERS signal intensity, greater overlap between groups was observed in the PCA analysis. The combined UC–SEC–UC approach resulted in the weakest signal, potentially due to partial loss of sEVs during the multiple isolation steps; however, PCA revealed a more uniform vesicle population. Therefore, the SEC–UC workflow was ultimately selected to minimise sEV loss while maintaining acceptable spectral quality and population consistency.

3.4 sEV Characterization

To confirm the presence of sEVs in the isolated samples, both physical and molecular characterization of nanostructures present was performed (Fig. 3). The size distribution and concentration of sEVs from healthy plasma were measured using NanoFCM. The isolated sEVs exhibited a mean particle size of 74 ± 1 nm and a concentration of approximately 1.34×10^{10} particles/mL (Fig. 3a). A TEM image showed a cup-shaped morphology of sEVs with sizes ranging from 70 to 100 nm (Fig. 3b). NanoFCM was further used to confirm sEV-specific protein markers. As shown in

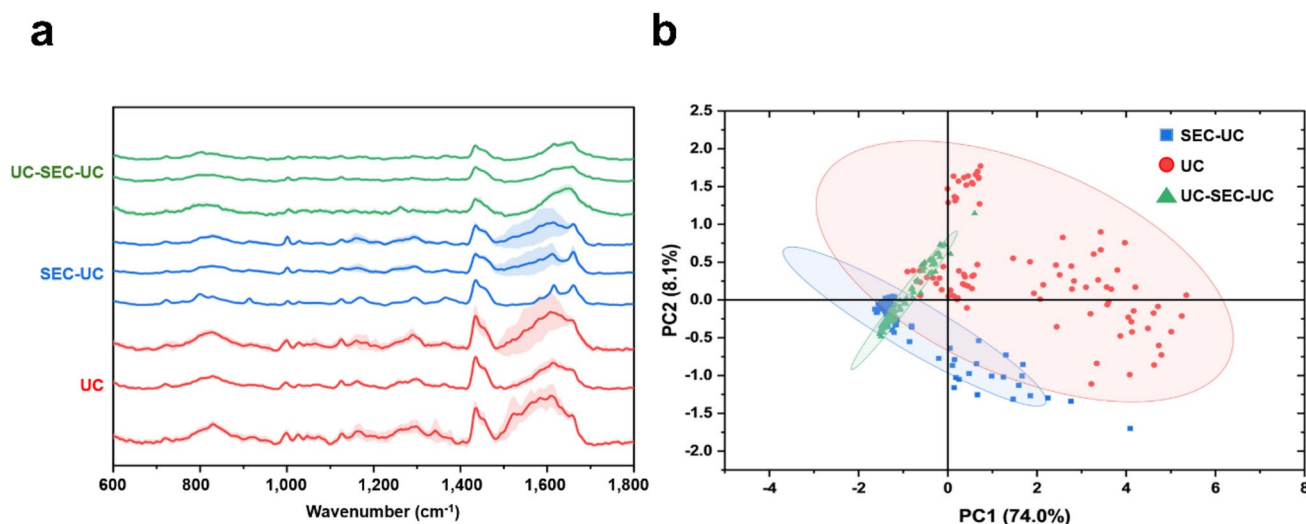


Fig. 2 Comparative SERS characterisation of sEVs isolated from healthy plasma by three isolation methods. **a** Mean normalised SERS spectra of sEVs isolated by UC alone (red), SEC followed by UC

(blue), and UC followed by SEC and UC (green). **b** PCA plot showing spectral separation between isolation methods, with PC1 (74.0%) and PC2 (8.1%) totally explaining 82.1% of total variance

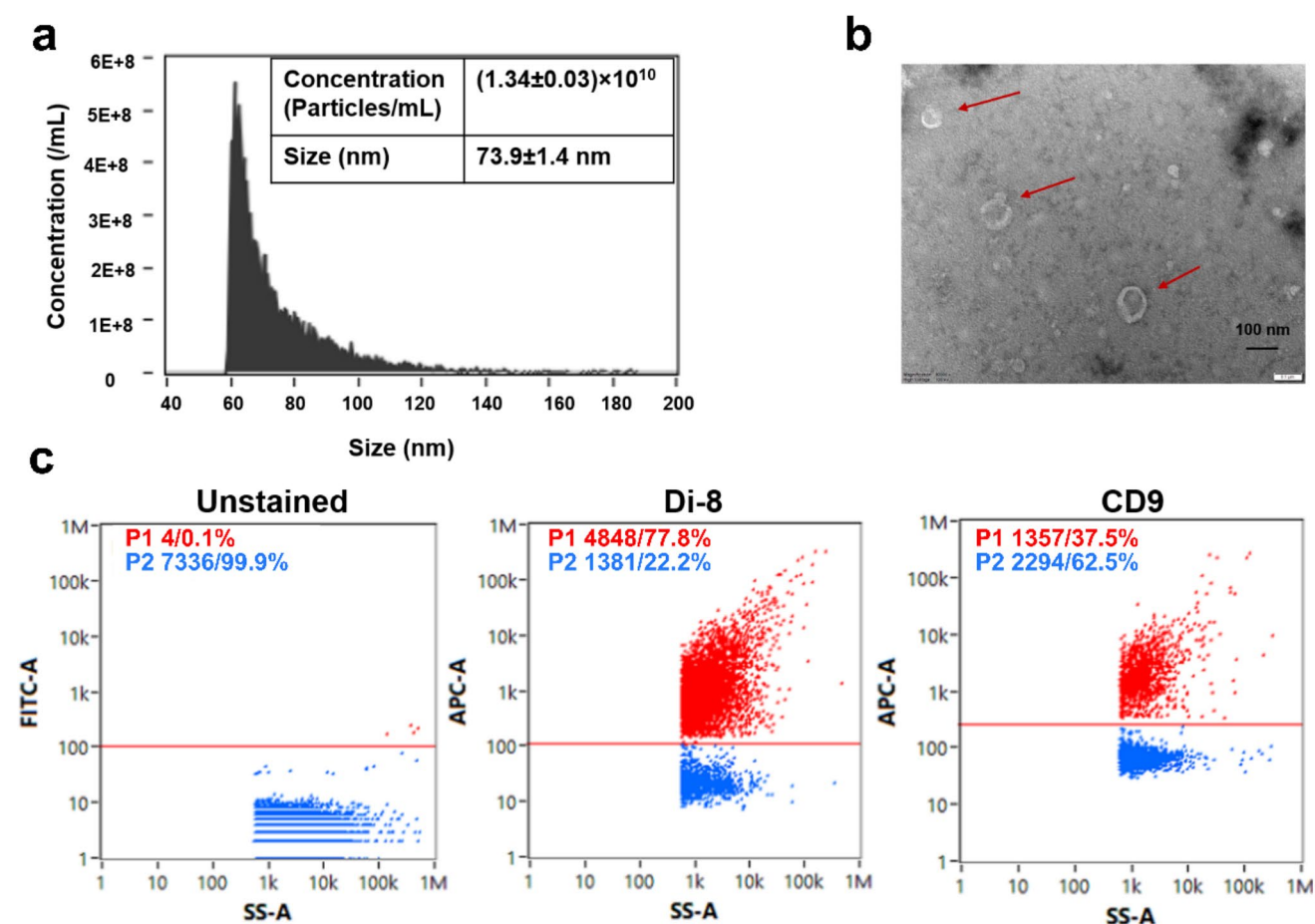


Fig. 3 Characterization of plasma-derived sEVs. **a** Size distribution and particle concentration of isolated sEVs measured using NanoFCM. **b** TEM image of isolated sEVs following negative staining. Images were acquired at an accelerating voltage of 100 kV and 80,000 $\times$ magnification, showing cup-shaped morphology of sEVs, with red arrows indi-

cating the presence of sEVs (scale bar = 100 nm). **c** Lipid dye labelling with Di-8 showed that 77.8% of detected particles were lipid-positive. Among these lipid-positive particles, 37.5% were CD9-positive, confirming successful isolation of sEVs as analysed by NanoFCM

Fig. 3c, staining with Di-8 lipid dye confirmed the presence of lipid membrane particles. Di-8 labelling demonstrated that approximately 78% of the isolated particles were lipid particles. Furthermore, labelling with an anti-CD9-Alexa 647 antibody showed that 37.5% of the particles expressed the EV-specific marker CD9, further supporting the presence of sEVs. These findings confirm both the successful isolation of sEVs and their presence in the samples included in the study.

3.5 SERS and PCA of Cancer Cell Line-Derived sEVs

To test the label-free SERS for breast cancer subtyping, three breast cell lines were selected as proof-of-concept demonstration: the non-tumorigenic breast epithelial cell line MCF-10A, the HER-2^{neg} breast cancer cell line MCF-7, and the HER-2^{pos} breast cancer cell line SKBR-3. This cell-line panel was used to assess whether label-free SERS could detect distinct molecular fingerprints in sEVs, and whether

these spectral differences were sufficient to separate healthy, HER-2^{neg}, and HER-2^{pos} phenotypes into defined clusters. It is important to note that the proof-of-concept experiments using cell lines and the clinical validation were performed on biologically distinct samples. Using single-cell-type models reduces the complexity of biological samples by eliminating the heterogeneity associated with multiple cellular origins, allowing the observed spectral differences to be attributed primarily to the cellular origin of the sEVs. The clinical study was performed to evaluate whether the discriminatory spectral features identified under controlled *in vitro* conditions could also be detected in patient-derived samples, thereby assessing the translational potential of the proposed approach. Therefore, the cell line model system enabled the identification of key discriminatory spectral features responsible for the observed separation. Conditioned cell culture media from each cell line were collected for sEV isolation. The samples were first concentrated by UC, purified by SEC, and subsequently reconcentrated by UC

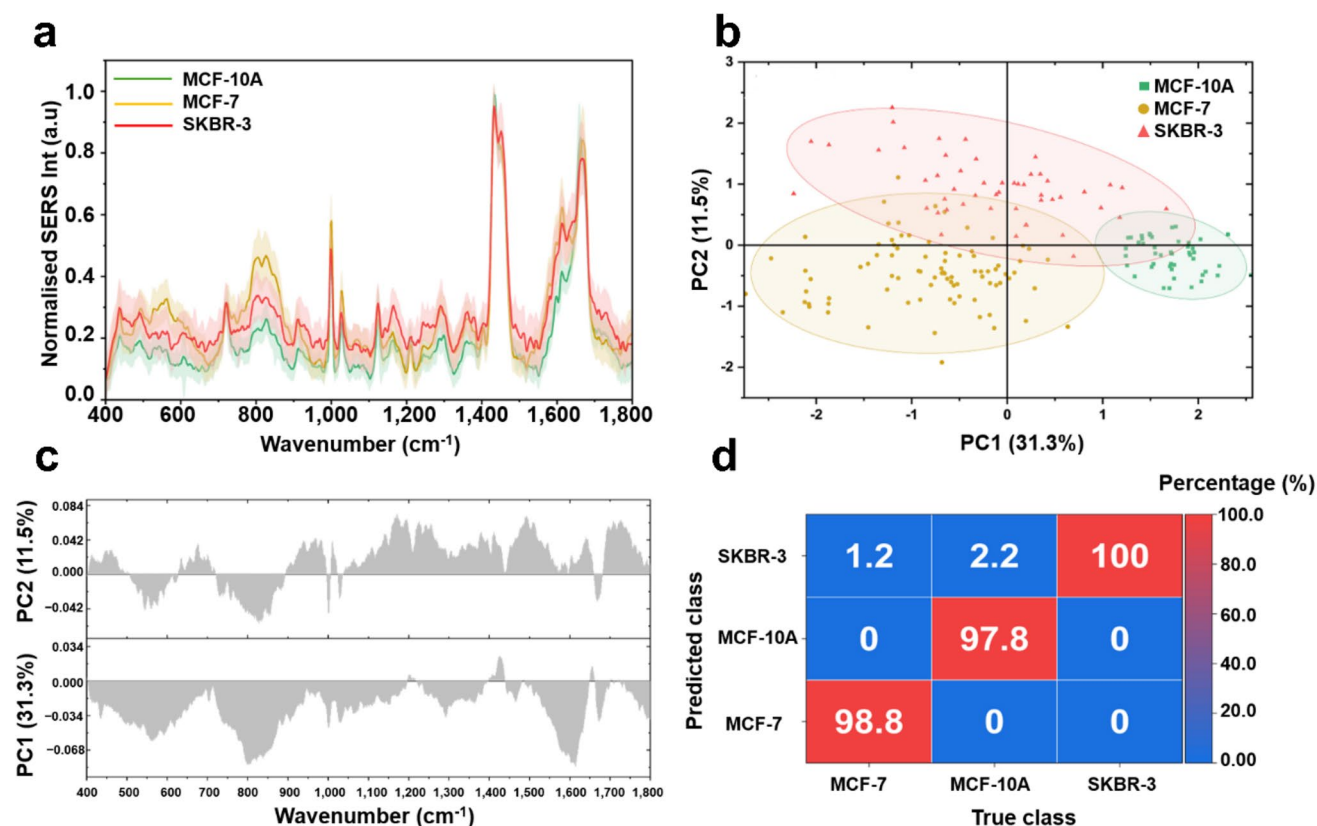


Fig. 4 SERS-based molecular profiling of sEVs from breast cancer and healthy breast cell lines. **a** Average SERS spectra of different cell line-derived sEVs. **b** PCA clustering analysis was performed to visualize separation among three groups. **c** Loading plots obtained from PCA analysis of MCF-10A, SKBR-3, and MCF-7-derived sEVs, indicating molecular regions in SERS spectra involved in group separation along

Table 1 Spectral regions and biomolecular assignments in sEV SERS spectra

Raman region	Assignment [28–30]
~ 540–580 cm ⁻¹	Low-wavenumber cellular vibrations; possible skeletal/deformation modes
~ 700–730 cm ⁻¹	Nucleic-acid/lipid-associated modes; adenine-related contributions can occur in this region
~ 780–790 cm ⁻¹	Nucleic-acid backbone/DNA/RNA-associated contributions
~ 1000–1010 cm ⁻¹	Phenylalanine ring breathing and protein-associated signal
~ 1260–1340 cm ⁻¹	Amide III/CH deformation/nucleic-acid and protein-associated bands
~ 1440–1470 cm ⁻¹	CH ₂ /CH ₃ deformation from lipids and proteins
~ 1640–1670 cm ⁻¹	Amide I/protein secondary-structure and lipid C=C contributions

prior to SERS analysis. The obtained spectra were analysed using PCA to evaluate natural clustering patterns and identify spectral features contributing to group separation, followed by SVM modelling for classification and prediction.

PC1 and PC2. **d** Confusion matrix of prediction results for SERS data-sets of MCF-10A, SKBR-3, and MCF-7 cell line-derived sEVs. Model demonstrated excellent classification performance, with accuracies of 97.8% for MCF-10A, 100% for SKBR-3, and 98.8% for MCF-7, indicating that three cell lines can be distinguished with high accuracy based on their SERS spectral signatures

As indicated in Fig. 4, SERS measurements were performed on 3 biological replicates. From each sample, we extracted 30 spectra, resulting in a total of 90 spectra per cell line. The SERS spectra (in the range of 400–1800 cm⁻¹) were collected, smoothed, baseline corrected, and then normalised. The averages of all collected spectra are presented in Fig. 4a.

sEVs from all three cell lines produced characteristic SERS spectra across the 400–1800 cm⁻¹ fingerprint region, consistent with the complex biomolecular composition of sEVs, including proteins (~ 1000–1010, ~ 1260–1340, and ~ 1600–1800 cm⁻¹), lipids (~ 1400–1470 cm⁻¹), and nucleic acids (~ 700–790 cm⁻¹) (Table 1). The spectral profiles of sEVs have slight variations, which are due to the differences in their molecular composition, as shown in Fig. 4a. We speculate that these differences are connected to inherited biomolecular profiles from their originating cells. It is difficult to distinguish these spectral variations using visual observations or by pinpointing separate peaks. Therefore, we have applied PCA and SVM analyses to cluster the

spectra into known groups and differentiate among healthy, HER-2^{neg}, and HER-2^{pos} cell line-derived sEVs.

Each spectrum was treated as a single observation, with Raman-shift intensities serving as input variables. PCA was performed on the SERS spectral dataset, and the first five principal components (PC1–PC5) were retained for downstream analysis to capture a broader proportion of spectral variance. For the cell-culture-derived sEV dataset, PC1 accounted for 31.28% of the total variance, followed by PC2 (11.54%), PC3 (6.76%), PC4 (3.90%), and PC5 (3.55%), together explaining 57.03% of the cumulative variance (Table S1). Although PC1 and PC2 were used for visualisation in the PCA score plots (Fig. 4b), given their comparatively higher explained variance, all five components were retained for subsequent classification analysis. SKBR-3 sEVs clustered predominantly in the upper-left quadrant, MCF-7 sEVs in the lower-central region, and MCF-10A sEVs toward the right along PC1 in Fig. 4b. Confidence ellipses showed partial overlap between groups, particularly between MCF-7 and MCF-10A along PC2, indicating that while spectral differences are present, within-group biological variability contributes to distributional spread. PCA loading plots (Fig. 4c) were generated from the component/eigenvector output to identify Raman-shift regions contributing to the observed separation. Spectral features with large positive or negative loading values were considered major contributors to variance along the corresponding principal component, and neighbouring high-loading variables were grouped into broader Raman bands for interpretation. PC loading plot indicates the key spectral features involved in discriminating between the sample groups. The loading plots of PC1 and PC2 reveal distinct spectral contributions across the fingerprint region (400–1800 cm⁻¹). More specifically, we have identified variability in the major biomolecular regions (Fig. 4c and Table 1). In terms of the spectral difference between cancerous and healthy cell lines, the most distinct features occurred in Amide I and III regions (1550–1650 cm⁻¹, 1250–1350 cm⁻¹), indicating the presence of membrane proteins with different structures. The spectral features on loading plots, such as the peaks in the PC1 panel, indicate that healthy cell-derived sEVs have more α -helical features (1650 cm⁻¹), while cancerous cell-derived sEVs have more β -sheet structures (Raman shift toward 1675 cm⁻¹ or broader bands spanning 1650–1680 cm⁻¹), and the same region can also indicate unfolded proteins. In the case of lipid vesicles, the proteins on or between the membranes are most likely to be α -helical or β -sheet structured. For cancer phenotypes MCF-7 (HER-2^{neg}) and SKBR-3 (HER-2^{pos}), the peaks in the PC2 panel indicate that most of the spectral distinction happens in regions such as ~ 800 – 900 , ~ 1000 – 1100 , and ~ 1500 – 1600 cm⁻¹, which

are usually assigned to changes in side chain groups on the aromatic ring of amino acids.

Although PCA was useful for exploratory spectral analysis, it did not achieve satisfactory clustering between the groups. Therefore, SVM analysis was applied to enhance group separation and improve classification performance. The resulting confusion matrix of the 5-fold stratified cross-validation check is shown in Fig. 4d. The SVM model achieved a recall of 98.8% for the HER-2^{neg} cell line (MCF-7), 100% for the HER-2^{pos} cell line (SKBR-3), and 97.8% for the healthy breast epithelial cell line (MCF-10A), demonstrating highly accurate classification among the three cell lines, indicating the potential utility of this method for sEV classification.

3.6 Machine Learning-Assisted SERS Analysis of Plasma-Derived sEVs for Breast Cancer Subtyping

In this study, 42 plasma samples were included, comprising 28 breast cancer patients and 14 healthy donors. sEVs were isolated from each sample using the SEC–UC method, followed by label-free SERS measurement. To improve spectral reliability and reproducibility, three technical replicates were obtained per sample, with 30 SERS spectra collected per replicate, resulting in a total of 90 spectra per plasma sample. The average normalised SERS spectra of the plasma-derived sEVs from each group are shown in Fig. 5a. For the sEVs derived from plasma of patients, the spectra show visible differences in the Amide I and Amide III regions, similar to those observed in cell-culture-derived sEVs. These changes may indicate differences in the protein structures carried by sEVs derived from healthy donors and cancer patients. Plasma-derived sEVs are a heterogeneous population originating from multiple cell types, and their molecular profiles may reflect a combination of tumor-associated and physiological changes. Therefore, the SERS signatures observed in this study represent the overall molecular characteristics of circulating sEV populations.

The SVM model was employed as a classifier because it is well suited to high-dimensional SERS datasets with a relatively small number of biological samples. It can separate complex, high-dimensional, and non-linear spectral patterns even when class differences are subtle. It also performs effectively with relatively small datasets by maximizing the separation margin, although careful cross-validation is essential to avoid overfitting [31]. The normalised spectra from these samples were used to develop an SVM model to classify three groups among healthy, HER-2^{neg}, and HER-2^{pos} breast cancer. For the model development, we used data acquired from ten HER-2^{neg} patients, ten HER-2^{pos} patients, and ten healthy donors.

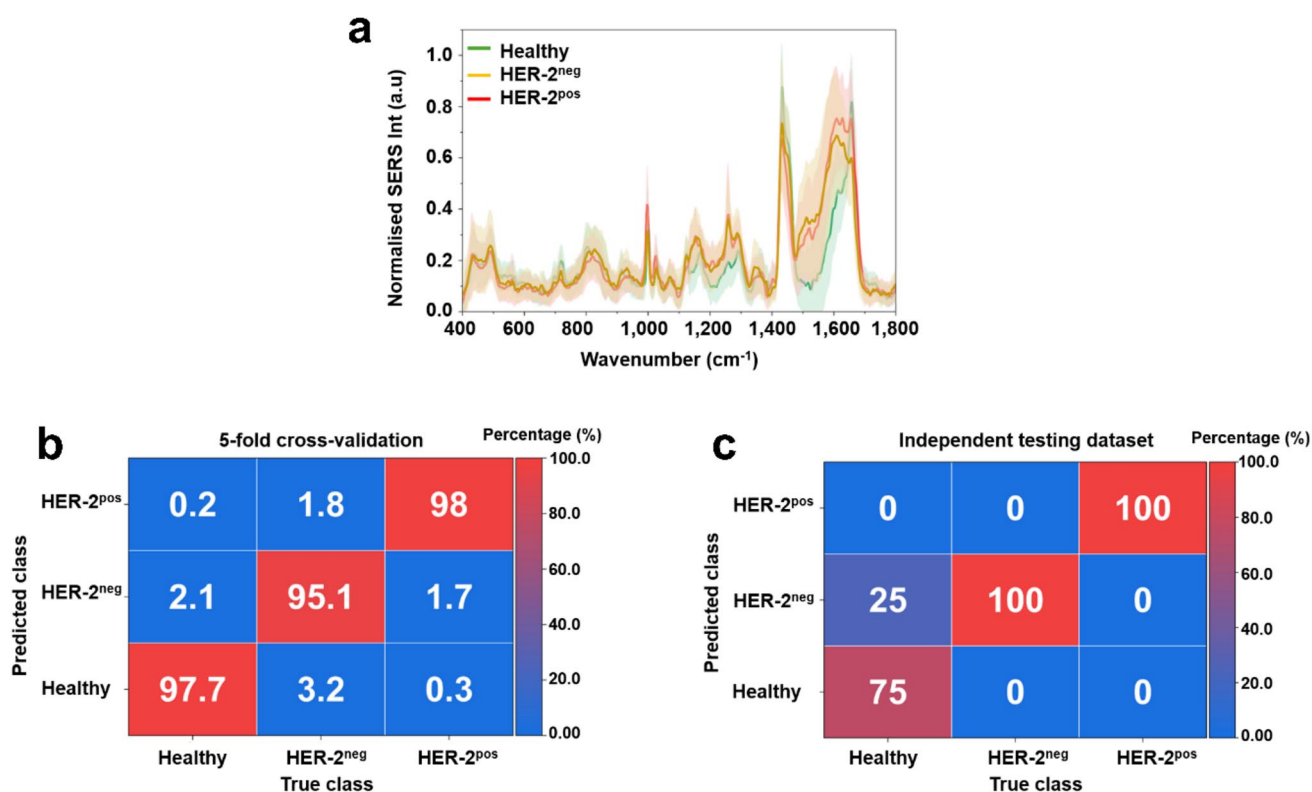


Fig. 5 SERS profiling of sEVs from breast cancer patients and healthy controls. **a** Average SERS spectra of plasma-derived sEVs. SERS spectra are smoothed, baseline corrected, and normalised. **b** Confusion matrix for classifying healthy, HER-2^{neg}, and HER-2^{pos} of pre-testing dataset used for training machine learning model (5-fold cross-validation). Model achieved high classification performance, with correct

classification of 97.7% for healthy, 95.1% for HER-2^{neg}, and 98% for HER-2^{pos} breast cancer samples. Overall accuracy was 96.9% for pre-testing dataset. **c** Confusion matrix of independent testing dataset used for validating SVM model achieved 100% accuracy for both HER-2^{neg} and HER-2^{pos} groups, and 75% accuracy for control group

Table 2 Classification performance of SERS-based SVM model on training dataset

Class	Precision (%)	Recall (%)	F1-score (%)
Healthy	97.7	96.7	97.2
HER-2 ^{neg}	95.1	96.1	95.6
HER-2 ^{pos}	98.0	98.0	98.0

To assess model performance and evaluate its translational applicability, two distinct validation strategies were employed: (i) a random intra-cohort 70/30 train-test split and (ii) an independent external test set validation. In the first approach, the dataset was randomly divided into 70% training data and 30% testing data. Validation using this method showed that the overall classification accuracy reached approximately 95%, with accuracies of 94.7% for healthy controls, 95.8% for HER-2^{neg} samples, and 96.4% for HER-2^{pos} samples (Fig. S3). This validation strategy has been widely reported in previous studies [19, 32]. In the second strategy, an independent validation cohort comprising samples from an additional four HER-2^{neg} patients, four HER-2^{pos} patients, and four healthy donors, which were not included in model training, was used to evaluate model performance on completely unseen patient samples.

Table 3 Classification performance of SERS-based SVM model on testing dataset

Class	Precision (%)	Recall (%)	F1-score (%)
Healthy	100.0	75.0	85.7
HER-2 ^{neg}	80.0	100.0	88.9
HER-2 ^{pos}	100.0	100.0	100.0

The confusion matrix of the training dataset demonstrated that the true positive rate (recall), representing the sensitivity of detection for healthy, HER-2^{neg}, and HER-2^{pos} groups, was 96.7%, 96.1%, and 98.0%, respectively (Table 2). In addition, the precision values for healthy, HER-2^{neg}, and HER-2^{pos} groups were 97.7%, 95.1%, and 98.0%, respectively (Fig. 5b). The overall classification accuracy was evaluated as 96.9%. For testing, an independent dataset was used to validate the model. The results demonstrated sensitivity values of 75% for healthy controls and 100% for both HER-2^{neg} and HER-2^{pos} breast cancer groups (Table 3), with an overall prediction accuracy of approximately 85% (Fig. 5c). We recommend independent cohort validation, as it provides more realistic and reliable estimates of model performance compared with internal cohort validation, which may overestimate accuracy due to shared

characteristics among samples originating from the same experimental cohort.

The relatively small sample size is a limitation of this study. However, the consistent SERS spectra obtained across three technical replicates (90 spectra per sample) and the reproducible measurements acquired at different times using a standardized protocol suggest that the observed spectral differences primarily reflect biological variation rather than random sampling variability. Nevertheless, validation in larger, independent cohorts is required to confirm the robustness and generalizability of the proposed approach in future studies.

Although the SERS-based classification model demonstrated promising diagnostic performance, the reduced specificity observed in the independent validation cohort, including the misclassification of some healthy controls as HER-2^{neg} breast cancer, highlights a limitation of this pilot study. In addition, the relatively small sample size ($n = 42$) may have limited the statistical power and generalizability of the findings. These observations underscore the need for larger training and independent validation cohorts to improve model robustness, enhance predictive performance, and support future clinical translation. Furthermore, samples included in this study were collected from multiple clinical centers, providing an initial assessment of the reproducibility and potential generalizability of the proposed model. Nevertheless, additional multicenter studies involving more diverse patient populations are required to further confirm the reproducibility and clinical utility of this approach.

Further research involving larger, multicenter cohorts with more diverse patient populations and independent external validation is needed to confirm the reproducibility, robustness, and clinical applicability of this approach. Further research should also consider incorporating additional clinical parameters, including inflammation, metabolic disorders, autoimmune diseases, and other physiological or pathological conditions, to develop more advanced models with improved robustness, specificity, and clinical applicability.

4 Conclusions

In this study, we demonstrated the feasibility of combining label-free SERS with an SVM-based machine learning model for the analysis of plasma-derived sEVs to diagnose breast cancer and subtype HER-2^{neg} and HER-2^{pos} tumors. sEVs isolated from plasma using a SEC-UC approach were directly analyzed by label-free SERS without prior biomarker selection or labelling. This approach enabled the acquisition of comprehensive molecular fingerprints

reflecting the overall biochemical composition of plasma-derived sEVs, including proteins, lipids, nucleic acids, and other biomolecules, rather than relying on specific sEV-associated biomarkers.

Our results also show that plasma-derived sEVs contain detectable SERS molecular fingerprints that reflect disease-associated biochemical differences. While PCA was useful for exploratory analysis, it provided only partial group separation; therefore, SVM was applied to improve classification performance. In the clinical cohort, the SVM model trained on these spectral signatures achieved an overall classification accuracy of approximately 95% during internal cross-validation and approximately 85% in independent external validation, while maintaining sensitivity of 100% for both breast cancer subtypes in the independent test set.

It should be pointed out that SERS spectra can be influenced by factors such as laser power, instrument calibration, acquisition parameters, and differences between Raman instruments. In the present study, all spectra were acquired using the same instrument under standardized acquisition conditions to minimize technical variability; however, the reproducibility and transferability of the method across different instruments and experimental settings remain to be established. Therefore, the findings should be considered a pilot proof-of-concept demonstrating the feasibility of combining SERS analysis of plasma-derived sEVs with machine learning for breast cancer classification. Future studies involving larger cohorts from additional independent clinical centers, more diverse patient populations, comprehensive clinical characterization, standardized acquisition protocols, and external validation across different instruments are needed to confirm the reproducibility, robustness, and clinical applicability of this approach.

In conclusion, this study demonstrates the feasibility of using label-free SERS profiling of plasma-derived sEVs, combined with SVM-based machine learning, as a minimally invasive strategy for breast cancer detection and molecular subtyping. The central clinical question addressed was whether biochemical signatures carried by circulating sEVs could distinguish healthy donors from breast cancer patients and, importantly, differentiate HER-2^{neg} from HER-2^{pos} breast cancer directly from plasma, without the need for tissue biopsy, antibody labelling, or pre-selection of specific biomarkers.

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Author Contribution Fatemeh Pashizeh and Anastasiia Tukova conducted the experiments, performed the data analysis, and wrote the manuscript. Victoria Le Ching Tan provided clinical insights regarding breast cancer patients. Simon Chang-Hao Tsao and Dhanusha Sabanathan facilitated access to breast cancer patient samples and provided clinical insights, with support from Su Su Thae Hnit and Yuling Wang, who supervised the project. All authors discussed the results and contributed to the final version of the manuscript.

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Data Availability The relevant data are available from the authors upon reasonable request.

Declarations

Conflicts of interest The authors declare that they have no conflicts of interest.

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